

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 322 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	King's Property & Definite Integral Properties	173 PYQs	Integral Properties
Very High	Limit of a Sum & Area Under Curves	150 PYQs	Summation & Area

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
King's Property & Definite Integral Properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, $\int_{-a}^a f(x) dx = 0$ (if odd), $\frac{d}{dx} \int u(x) v(x) f(t) dt = f(v(x)) v'(x) - f(u(x)) u'(x)$
Limit of a Sum & Area Under Curves	$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$, $\text{Area between } y^2 = 4ax \text{ and } y = mx \text{ implies } \frac{8a^2}{3m^3}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 King's Property & Definite Integral Properties (Very High Yield — 173 PYQs)

Core Principle: King's property $f(a+b-x)$, reflection symmetry, integrals of even/odd functions, periodic functions, and Leibnitz integral differentiation.

■■ **Caution / JEE Pitfall:** King's property added to the original integral often yields $2I = \int_a^b 1 dx = b - a \implies I = (b - a) / 2$.

■ **Speed Trick:** Integral of odd function over symmetric limits $[-a, a]$ is always ZERO: $\int_{-a}^a (-a)^a f(x) dx = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Let $f(x) = \frac{1}{3}$, $x \leq \pi/2$ and $f(x) = \frac{1}{3} \sin(x)$, $x > \pi/2$. If f is continuous a..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $(2^{1-a} + 2^{1+a}), f(a), (3^a + 3^{-a})$ be in A.P. and α be the minimum value of $f(a)$. Then the value of the integral $\int_{\alpha}^{\dots} \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "The integral $\int_0^1 \cot^{-1}(1+x+x^2) dx$ is equal to ..." [Ans: N/A]

3.2 Limit of a Sum & Area Under Curves (Very High Yield — 150 PYQs)

Core Principle: Definite integrals as limits of Riemann sums ($\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$), and enclosed area calculations between curves.

■ **Caution / JEE Pitfall:** Area is always a positive geometric quantity; if the curve dips below the x-axis, take absolute value $|\int y dx|$.

■ **Speed Trick:** Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is given directly by $\frac{16ab}{3}$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}(\sqrt{x+1})}{(\sqrt{x^4+x^2+1})} dx$ is equal to:..." [Ans: N/A]
- **[2026 Morning Shift] Q#1 (MCQ):** "The value of the integral $\int_{-(\pi/4)}^{(\pi/4)} (\frac{3}{2} \cos^4 x + e^{\sin x}) dx$ is :..." [Ans: B]
- **[2026 Evening Shift] Q#1 (MCQ):** "If $\alpha=1$ and $\beta=1+i\sqrt{2}$, where $i=\sqrt{-1}$ are two roots of the equation $x^3+ax^2+bx+c=0, a, b, c \in \mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \alpha^k$..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The value of the integral $\int_{-(\pi/4)}^{(\pi/4)} (\frac{3}{2} \cos^4 x + e^{\sin x}) dx$ is :

- (A) $4\pi+2$
- (B) $3\pi+8$ [Correct]
- (C) $3\pi+4$
- (D) $4\pi+3$

Correct Answer: B

Solution Analysis: Let $I = \int_{-(\pi/4)}^{(\pi/4)} (\frac{3}{2} \cos^4 x + e^{\sin x}) dx$. We will use the standard property: If $I = \int_{-a}^a f(x) dx$, then $I = \int_{-a}^a f(-x) dx$ and hence $2I = \int_{-a}^a (f(x) + f(-x)) dx$. Now here, $f(x) = \frac{3}{2} \cos^4 x + e^{\sin x}$. So, $f(-x) = \frac{3}{2} \cos^4(-x) + e^{\sin(-x)} = \frac{3}{2} \cos^4 x + e^{-\sin x}$. Using $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, $f(-x) = \frac{3}{2} \cos^4 x + e^{-\sin x}$. Therefore, $f(x) + f(-x) = \frac{3}{2} \cos^4 x (1 + 1) + (e^{\sin x} + e^{-\sin x}) = 3 \cos^4 x + (e^{\sin x} + e^{-\sin x})$. Now simplify the bracket: $e^{\sin x} + e^{-\sin x} = \frac{e^{\sin x} + e^{-\sin x}}{1}$. Let $t = e^{\sin x}$. Then $e^{-\sin x} = \frac{1}{t}$. So, $\frac{e^{\sin x} + e^{-\sin x}}{1} = \frac{t + \frac{1}{t}}{1} = \frac{t^2 + 1}{t}$.

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$, $x \in \mathbb{R}$ and let

$g(x) = \int_0^x (f(t) + 2)^{15} (t-4)^6 (t+12)^{17} dt$, $x \in \mathbb{R}$. If p and q are respectively the points of local minima and local maxima of g , then the value of $|p+q|$ is equal to _____.

Correct Answer: 9

Solution Analysis: The given equation is: $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$. Since $e^{x-t} = e^x \cdot e^{-t}$, we can move e^x outside the integral because it is constant relative to t : $f(x) = 1 - 2x + e^x \int_0^x e^{-t} f(t) dt$. Differentiate both sides with respect to x using the Leibniz Rule and the product rule: $f'(x) = -2 + [e^x \int_0^x e^{-t} f(t) dt + e^x (e^{-x} f(x))]$. Now, we use substitution from our original equation. Notice that $e^x \int_0^x e^{-t} f(t) dt = f(x) - (1 - 2x)$. Substituting this back into the derivative expression: $f'(x) = -2 + [f(x) - (1 - 2x) + f(x)] \Rightarrow f'(x) = 2$.